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① $\bar{X} = 52 \quad s = 10 \quad n = 100$

a) $SE = \frac{\sigma}{\sqrt{n}}$

$\frac{10}{\sqrt{100}} = \frac{10}{10} = 1$

b) 95% CI = $(\hat{\beta} - c \cdot SE \ , \ \hat{\beta} + c \cdot SE) = (52 - 1.96(1) \ , \ 52 + 1.96(1)) = (50.04, 53.96)$ → Reject null H_0 that $\mu = 50$ at 95% CI b/c 50 not in interval

Z stat: $\frac{\hat{\beta} - H_0}{SE} = \frac{52 - 50}{1} = 2$ } 2 > 1.645 & 1.96 so reject null $H_0: \mu = 50$ at 90% & 95% CI (0.10 & 0.05 α).
2 < 2.576 so fail to reject null at 99% CI & 1% significance level

CI: * double check

90% = $52 \pm 1.645(1) = (50.355, 53.645)$ → Reject null H_0 that $\mu = 50$ at 90% CI b/c 50 not in interval

99% = $52 \pm 2.576(1) = (49.424, 54.576)$ → Fail to reject null at 99% CI & 1% significance level b/c 50 in interval

② $\hat{S} = 3.1 \quad \widehat{SE}(\hat{S}) = 1.3$

90%: $(3.1 - 1.645(1.3), 3.1 + 1.645(1.3)) = (0.9615, 5.2385)$

95%: $(3.1 - 1.96(1.3), 3.1 + 1.96(1.3)) = (0.552, 5.648)$

99%: $(3.1 - 2.576(1.3), 3.1 + 2.576(1.3)) = (-0.2488, 6.4488)$

We reject the null at the 90% CI and 10% significance level ($\alpha = 0.10$) and at the 95% CI and 5% significance level ($\alpha = 0.05$) because neither of these intervals contain 0. We fail to reject the null of $S = 0$ at the 99% CI and 1% significance level ($\alpha = 0.01$) because the interval includes 0, meaning the data isn't strong enough to rule the null out at this level.

③ $\bar{X} = 11.5 \quad s = 3 \quad n = 36 \quad , H_0: \mu = 12 \quad , H_A: \mu \neq 12$

$SE: \frac{3}{\sqrt{36}} = \frac{3}{6} = 0.5$ CI: $(11.5 - 1.96(.5), 11.5 + 1.96(.5)) = (10.52, 12.48)$

→ we fail to reject the null that $\mu = 12$ because 12 is present in the 95% CI, meaning it is a plausible value for the mean

$Z = \frac{(\bar{X}_A - \bar{X}_B - \text{null})}{SE_{\text{difference}}}$

④ $\bar{X}_A = 25 \quad S_A = 8 \quad n_A = 100 \quad \bar{X}_B = 22 \quad S_B = 7 \quad n_B = 100 \quad H_0: \mu_A - \mu_B = 0 \quad H_A: \mu_A \neq \mu_B$

Var = σ^2/n

$SE_{(\mu_A - \mu_B)} = \sqrt{\left(\frac{S_A^2}{n_A} + \frac{S_B^2}{n_B}\right)}$

$SE_A = \frac{8}{\sqrt{100}} = 0.8 \quad SE_B = \frac{7}{\sqrt{100}} = 0.7$

$SE(\mu_A - \mu_B) = \sqrt{\frac{8^2 + 7^2}{100}} \approx 1.063$

$Z = \frac{25 - 22 - 0}{1.063} = 2.822$ } 2.822 > 1.645, 1.96, 2.576 so we reject null of $\mu_A = \mu_B$ (or $\mu_A - \mu_B = 0$) because Z stat is larger than all critical values

CI: $\bar{X}_A - \bar{X}_B \pm c \cdot SE_{\text{diff}}$

$3 \pm 1.96(1.063)$ } 95% = (0.91652, 5.08348)
↳ reject the null @ the 95% CI / 5% stat. sig. level because 0 is not in the interval

If we repeat the sample & compute procedure many times, ~95% of the intervals would contain the true value $\mu_A - \mu_B$ of the difference in means

⑤ $SE_{(\mu_A - \mu_B)} = \sqrt{\frac{6^2}{81} + \frac{5^2}{64}} \approx 0.914$

$Z = \frac{14 - 12 - 0}{0.914} \approx 2.188$ → FTR @ 99% CI & 1% Sig. lvl

Reject @ 90%, 95% CI & 5%, 10% Sig. lvl

⑥ $SE_{(\mu_A - \mu_B)} = \sqrt{\frac{1.5^2}{49} + \frac{1.4^2}{64}} \approx 0.277$

$Z = \frac{3.2 - 2.8}{0.277} \approx 1.444$

95% CI = $0.4 \pm 1.96(0.277) = (-0.1429, 0.9429)$

The difference is not statistically large relevant to the noise because the z stat is less than the critical value at every α level. We Fail to Reject $H_0: \mu_A - \mu_B = 0$ at the 95% CI because 0 is included in the interval. We are 95% confident that the diff in means of factories A & B is between (-0.1429, 0.9429).

★ QUESTIONS 7 - 10 IN .IPYNB FILE IN THIS REPO ★